United States Patent	12400638
Kind Code	B2
Date of Patent	August 26, 2025
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Using aligned text and speech representations to train automatic speech recognition models without transcribed speech data

Abstract

A method includes receiving training data that includes unspoken textual utterances in a target language. Each unspoken textual utterance not paired with any corresponding spoken utterance of non-synthetic speech. The method also includes generating a corresponding alignment output for each unspoken textual utterance using an alignment model trained on transcribed speech utterance in one or more training languages each different than the target language. The method also includes generating a corresponding encoded textual representation for each alignment output using a text encoder and training a speech recognition model on the encoded textual representations generated for the alignment outputs. Training the speech recognition model teaches the speech recognition model to learn how to recognize speech in the target language.

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Appl. No.: **18/355508**

Filed: **July 20, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240029715 A1	Jan. 25, 2024

Related U.S. Application Data

us-provisional-application US 63369213 20220722

Publication Classification

Int. Cl.: **G10L15/06** (20130101)

U.S. Cl.:

CPC **G10L15/063** (20130101);

Field of Classification Search

CPC: G10L (15/063); G10L (15/16); G10L (13/08); G10L (15/26); G06N (3/044)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This U.S. patent application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application 63/369,213, filed on Jul. 22, 2022. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to using aligned text and speech representations to train automatic speech recognition models without transcribed speech data.

BACKGROUND

(2) Automatic speech recognition (ASR), the process of taking an audio input and transcribing it into text, has greatly been an important technology that is used in mobile devices and other devices. In general, automatic speech recognition attempts to provide accurate transcriptions of what a person has said by taking an audio input (e.g., speech utterance) and transcribing the audio input into text. Modern ASR models continue to improve in both accuracy (e.g. a low word error rate (WER)) and latency (e.g., delay between the user speaking and the transcription) based on the ongoing development of deep neural networks. However, one challenge in developing deep learning-based ASR models is a substation amount of transcribed speech is required during training. In some instances, unspoken text data is used to train the ASR models to supplement a small set of transcribed speech training data. Yet, this challenge is further complicated when training ASR models with low-resource languages that include zero available transcribed speech training data.

SUMMARY

(3) One aspect of the disclosure provides a computer-implemented method that when executed on data processing hardware causes the data processing hardware to perform operations for using aligned text and speech representations to train automatic speech recognition models without transcribed speech data. The operations include receiving training data that includes unspoken textual utterances in a target language. Each unspoken textual utterance not paired with any corresponding spoken utterance of non-synthetic speech. The operations also include generating a corresponding alignment output for each unspoken textual utterance of the received training data using an alignment model trained on transcribed speech utterances in one or more training languages each different than the target language. The operations also include generating a corresponding encoded textual representation for each alignment output using a text encoder. The operations also include training a speech recognition model on the encoded textual representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language to teach the speech recognition model to learn how to recognize speech in the target language.

(4) Implementations of the disclosure may include one or more of the following optional features. In some implementations, training the speech recognition model includes training the speech recognition model without using any transcribed speech utterances in the target language for supervised learning. In some examples, the speech recognition model includes an audio encoder and a decoder. The decoder may include a recurrent neural network transducer (RNN-T) architecture. In these examples, the audio encoder includes the text encoder, a speech encoder, and a shared encoder. Here, the encoder may include a plurality of multi-headed self-attention layers. The audio encoder may include a Conformer encoder. In these examples, the operations further include conditioning at least one of the audio encoder or the decoder on a language identifier uniquely identifying the target language. Conditioning the at least one of the audio encoder or the decoder includes conditioning the at least one of the audio encoder or the decoder on the language identifier using residual adaptor layers. The alignment model may be trained on transcribed speech utterances in one or more training languages different than the target language.

(5) In some implementations training the speech recognition model includes: generating, using a

shared encoder, a first encoded shared representation of the alignment output in a shared latent representation space for each alignment output; for each transcribed speech utterance in the one or more training languages, determining an encoded audio representation of the transcribed speech utterance using a speech encoder and generating a second encoded shared representation of the transcribed speech utterance in the shared latent representation space. Here, training the speech recognition model includes training the speech recognition model on the first encoded shared representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language and the second encoded shared representation generated for the transcribed speech utterances in the one or more training languages. Each unspoken textual utterance may include a sequence of words, word-pieces, graphemes, and/or phonemes. In some examples, the operations further include converting a script of the unspoken textual utterances in the target language into a phonetic representation shared across multiple languages using a pronunciation model.

(6) Another aspect of the disclosure provides a system that includes data processing hardware and memory hardware storing instructions that when executed on the data processing hardware causes the data processing hardware to perform operations. The operations include receiving training data that includes unspoken textual utterances in a target language. Each unspoken textual utterance not paired with any corresponding spoken utterance of non-synthetic speech. The operations also include generating a corresponding alignment output for each unspoken textual utterance of the received training data using an alignment model trained on transcribed speech utterances in one or more training languages each different than the target language. The operations also include generating a corresponding encoded textual representation for each alignment output using a text encoder. The operations also include training a speech recognition model on the encoded textual representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language to teach the speech recognition model to learn how to recognize speech in the target language.

(7) Implementations of the disclosure may include one or more of the following optional features. In some implementations, training the speech recognition model includes training the speech recognition model without using any transcribed speech utterances in the target language for supervised learning. In some examples, the speech recognition model includes an audio encoder and a decoder. The decoder may include a recurrent neural network transducer (RNN-T) architecture. In these examples, the audio encoder includes the text encoder, a speech encoder, and a shared encoder. Here, the encoder may include a plurality of multi-headed self-attention layers. The audio encoder may include a Conformer encoder. In these examples, the operations further include conditioning at least one of the audio encoder or the decoder on a language identifier uniquely identifying the target language. Conditioning the at least one of the audio encoder or the decoder includes conditioning the at least one of the audio encoder or the decoder on the language identifier using residual adaptor layers. The alignment model may be trained on transcribed speech utterances in one or more training languages different than the target language.

(8) In some implementations training the speech recognition model includes: generating, using a shared encoder, a first encoded shared representation of the alignment output in a shared latent representation space for each alignment output; for each transcribed speech utterance in the one or more training languages, determining an encoded audio representation of the transcribed speech utterance using a speech encoder and generating a second encoded shared representation of the transcribed speech utterance in the shared latent representation space. Here, training the speech recognition model includes training the speech recognition model on the first encoded shared representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language and the second encoded shared representation generated for the transcribed speech utterances in the one or more training languages. Each unspoken textual utterance may include a sequence of words, word-pieces, graphemes, and/or phonemes. In some

examples, the operations further include converting a script of the unspoken textual utterances in the target language into a phonetic representation shared across multiple languages using a pronunciation model.

(9) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic view of an example speech recognition system.

(2) FIG. **2** is a schematic view of an example speech recognition model.

(3) FIGS. **3**A-**3**C are schematic views of an example training process for training an audio encoder of the speech recognition model of FIG. **2**.

(4) FIG. **4** is a schematic view of an alignment model used during the example training process for training the audio encoder of the speech recognition model in FIGS. **3**A-**3**C.

(5) FIG. **5** is a schematic view of an example training process for training a duration predictor of the alignment model.

(6) FIG. **6** is a schematic view of an example training process for training an upsampler of the alignment model.

(7) FIG. **7** is a flowchart of an example arrangement of operations for a method of using aligned text and speech representations to train speech recognition models without transcribed speech data.

(8) FIG. **8** is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

(9) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(10) Training state-of-the-art automated speech recognition (ASR) models typically requires a substantial amount of labeled training data including speech utterances each paired with a corresponding transcription (i.e., ground-truth label). Obtaining the substantial amount of labeled training data can be very costly, especially for low-resource languages. Thus, recent approaches for training ASR models include self-supervised training using large amounts of unlabeled training data (i.e., speech not paired with any corresponding transcriptions and/or unspoken text not paired with any corresponding speech) to complement a relatively small amount of labeled training data. However, these approaches still require some small amount of labeled training data in a particular language the ASR model is being trained to recognize. Yet, in some instances, there is zero available labeled training data for certain low-resource languages. As such, in these instances, there is not any labeled training data to complement the large amounts of unlabeled training data.

(11) Accordingly, implementations herein are directed towards methods and systems of using aligned text and speech representation to train ASR models without transcribed speech training data. In particular, training the ASR model includes receiving training data that includes unspoken textual utterances in a target language where each unspoken textual utterance is not paired with any corresponding utterance of non-synthetic (or synthetic) speech. An alignment model generates a corresponding alignment output (i.e., aligned text representation) for each unspoken textual utterance of the received training data. Notably, the alignment model is trained on transcribed (i.e., labeled) speech utterances in one or more training languages each of which is different than the target language the ASR model is trained to recognize. Moreover, a text encoder generates a corresponding encoded textual representation for each alignment output. Thereafter, the ASR model is trained on the encoded textual representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language to teach the ASR model to

learn how to recognize speech in the target language when no labeled training data in the target language is available. That is, the ASR model is trained without using any transcribed speech utterances in the target language for supervised learning.

(12) FIG. **1** illustrates an ASR system **100** implementing an ASR model **200** that resides on a user device **102** of a user **104** and/or on a remote computing device **201** (e.g., one or more servers of a distributed system executing in a cloud-computing environment) in communication with the user device **102**. Although the user device **102** is depicted as a mobile computing device (e.g., a smart phone), the user device **102** may correspond to any type of computing device such as, without limitation, a tablet device, a laptop/desktop computer, a wearable device, a digital assistant device, a smart speaker/display, a smart appliance, an automotive infotainment system, or an Internet-of-Things (IoT) device, and is equipped with data processing hardware and memory hardware **113**.

(13) The user device **102** includes an audio subsystem **108** configured to receive an utterance **106** spoken by the user **104** (e.g., the user device **102** may include one or more microphones for recording the spoken utterance **106**) and convert the utterance **106** into a corresponding digital format associated with input acoustic frames **110** capable of being processed by the ASR system **100**. In the example shown, the user speaks a respective utterance **106** in a natural language of English for the phrase "What is the weather in New York City?" and the audio subsystem **108** converts the utterance **106** into corresponding acoustic frames **110** for input to the ASR system **100**. Thereafter, the ASR model **200** receives, as input, the acoustic frames **110** corresponding to the utterance **106**, and generates/predicts, as output, a corresponding transcription **120** (e.g., recognition result/hypothesis) of the utterance **106**. In the example shown, the user device **102** and/or the remote computing device **201** also executes a user interface generator **107** configured to present a representation of the transcription **120** of the utterance **106** to the user **104** of the user device **102**. In some configurations, the transcription **120** output from the ASR system **100** is processed, e.g., by a natural language understanding (NLU) module executing on the user device **102** or the remote computing device **201**, to execute a user command. Additionally or alternatively, a text-to-speech system (e.g., executing on any combination of the user device **102** or the remote computing device **201**) may convert the transcription into synthesized speech for audible output by another device. For instance, the original utterance **106** may correspond to a message the user **104** is sending to a friend in which the transcription **120** is converted to synthesized speech for audible output to the friend to listen to the message conveyed in the original utterance **106**.

(14) The ASR model **200** may operate in a streaming fashion, a non-streaming fashion, or some combination thereof. The ASR model **200** operates in the streaming fashion by, while receiving the sequence of acoustic frames **110**, encoding the sequence of acoustic frames **110** and then decoding the encoded sequence of acoustic frames **110** into an initial transcription (e.g., speech recognition result/hypothesis) **120**. Thus, the initial transcription **120** may correspond to words, word pieces, and/or individual characters generated by the ASR model **200** as soon as they are spoken. On the other hand, the ASR model **200** operates in the non-streaming fashion by receiving and processing additional right-context to improve upon the initial transcription **120** thereby generating a final transcription **120**. That is, the ASR model **200** processes additional input audio data or encoded acoustic frames (e.g., right-context) to improve the transcription **120** output by the ASR model **200**, but at increased latency.

(15) Referring to FIG. **2**, an example ASR model **200** may include a Recurrent Neural Network-Transducer (RNN-T) model architecture which adheres to latency constraints with interactive applications. The use of the RNN-T model architecture is exemplary only, as the ASR model **200** may include other architectures such as transformer-transducer and conformer-transducer model architectures among others. The RNN-T model **200** provides a small computational footprint and utilizes less memory requirements than conventional ASR architectures, making the RNN-T model architecture suitable for performing speech recognition entirely on the user device **102** (e.g., no communication with a remote server is required). The RNN-T model **200** includes an encoder

network **210**, a prediction network **220**, and a joint network **230**. The encoder network **210**, which is roughly analogous to an acoustic model (AM) in a traditional ASR system, includes a stack of self-attention layers (e.g., Conformer or Transformer layers) or a recurrent network of stacked Long Short-Term Memory (LSTM) layers. For instance, the encoder network (e.g., audio encoder) **210** reads a sequence of d-dimensional feature vectors (e.g., acoustic frames **110** (FIG. **1**)) x=(x.sub.1, x.sub.2, . . . , x.sub.T), where x.sub.t∈custom character.sub.d, and produces at each output step a higher-order feature representation. This higher-order feature representation is denoted as h.sub.1.sup.enc, . . . , h.sub.T.sup.enc. In some examples, the encoder network **210** includes a dual encoder framework that has a text encoder **202** and a speech encoder **204** (FIGS. **3**B and **3**C).

(16) Similarly, the prediction network **220** is also an LSTM network, which, like a language model (LM), processes the sequence of non-blank symbols output by a final Softmax layer **240** so far, y.sub.0, . . . , y.sub.ui−1, into a dense representation Putt. Together, the prediction network **220** and the joint network **230** may be referred to as a decoder that includes an RNN-T architecture. Finally, with the RNN-T model architecture, the representations produced by the encoder and prediction/decoder networks **210**, **220** are combined by the joint network **230**. The prediction network **220** may be replaced by an embedding look-up table to improve latency by outputting looked-up sparse embeddings in lieu of processing dense representations. The joint network then predicts P(y.sub.i|x.sub.t.sub.i, y.sub.0, . . . , y.sub.u.sub.i−1), which is a distribution over the next output symbol. Stated differently, the joint network **230** generates, at each output step (e.g., time step), a probability distribution over possible speech recognition hypotheses. Here, the “possible speech recognition hypotheses” correspond to a set of output labels each representing a symbol/character in a specified natural language. For example, when the natural language is English, the set of output labels may include twenty-seven (27) symbols, e.g., one label for each of the 26-letters in the English alphabet and one label designating a space. Accordingly, the joint network **230** may output a set of values indicative of the likelihood of occurrence of each of a predetermined set of output labels. This set of values can be a vector and can indicate a probability distribution over the set of output labels. In some cases, the output labels are graphemes (e.g., individual characters, and potentially punctuation and other symbols), but the set of output labels is not so limited. For example, the set of output labels can include wordpieces, phonemes, and/or entire words, in addition to or instead of graphemes. The output distribution of the joint network **230** can include a posterior probability value for each of the different output labels. Thus, if there are 100 different output labels representing different graphemes or other symbols, the output y.sub.i of the joint network **230** can include 100 different probability values, one for each output label. The probability distribution can then be used to select and assign scores to candidate orthographic elements (e.g., graphemes, wordpieces, and/or words) in a beam search process (e.g., by the Softmax layer **240**) for determining the transcription **120**.

(17) The Softmax layer **240** may employ any technique to select the output label/symbol with the highest probability in the distribution as the next output symbol predicted by the RNN-T model **200** at the corresponding output step. In this manner, the RNN-T model **200** does not make a conditional independence assumption, rather the prediction of each symbol is conditioned not only on the acoustics but also on the sequence of labels output so far. The RNN-T model **200** does assume an output symbol is independent of future acoustic frames **110**, which allows the RNN-T model to be employed in the streaming fashion, the non-streaming fashion, or some combination thereof.

(18) In some examples, the audio encoder **210** of the RNN-T model includes a plurality of multi-head (e.g., **8** heads) self-attention layers. For example, the plurality of multi-head self-attention layers may include Conformer layers (e.g., Conformer-encoder), transformer layers, performer layers, convolution layers (including lightweight convolution layers), or any other type of multi-head self-attention layers. The plurality of multi-head self-attention layers may include any number of layers, for instance 16 layers. Moreover, the audio encoder **210** may operate in the streaming

fashion (e.g., the audio encoder **210** outputs initial higher-order feature representations as soon as they are generated), in the non-streaming fashion (e.g., the audio encoder **210** outputs subsequent higher-order feature representations by processing additional right-context to improve initial higher-order feature representations), or in a combination of both the streaming and non-streaming fashion.

(19) FIGS. **3**A-**3**C illustrate an example training process **300** for training the ASR model **200** (FIG. **2**). The training process **300** described herein describes training the audio encoder **210** of the ASR model **200**, however, it is understood that the training process **300** may also include pre-training and/or fine-tune training of the audio encoder **210**. Moreover, implementations described herein contemplate the training process **300** training the audio encoder **210** of the ASR model **200** without training the decoder (e.g., prediction network **220** and joint network **230**) of the ASR model **200**. Yet, it is understood that the training process **300** may additionally or alternatively train other components of the ASR model **200** (e.g., prediction network **220** and/or joint network **230**) jointly with the audio encoder **210**.

(20) The training process **300** trains the audio encoder **210** using available training data that includes a set of unspoken textual utterances (X.sub.text) **320**, a set of transcribed non-synthetic speech utterances (X.sub.sup) **304**, and/or un-transcribed non-synthetic speech utterances (X.sub.unsup) **306**. Notably, each unspoken textual utterance **320** in the set of unspoken textual utterances **320** includes text-only data (i.e., unpaired data) in a target language such that each unspoken textual utterance **320** is not paired with any corresponding spoken audio representation (i.e., speech) of the utterance. Here, the target language is any language the training process **300** uses to train the audio encoder **210** to recognize where zero transcribed (i.e., labeled) training data is used during training. The unspoken textual utterance **320** may include any sequence text chunks including words, word-pieces, phonemes, and/or graphemes. Optionally, the available training data may also include the un-transcribed non-synthetic speech utterances **306** (also referred to as simply “un-transcribed speech utterance **306**”) each including audio-only data (i.e., unpaired data) in the target language such that the un-transcribed speech utterance **306** is not paired with any corresponding transcription. Notably, when the training data includes the un-transcribed speech utterances **306** in addition to the unspoken textual utterances **306**, the un-transcribed speech utterances **306** represent different training utterances than the unspoken textual utterances **320** such that the training process **300** cannot simply pair the unspoken textual utterances **320** with the un-transcribed speech utterances **306** to generate labeled training data.

(21) On the other hand, each transcribed non-synthetic speech utterance **304** (also referred to as simply “transcribed speech utterance **304**”) includes a corresponding transcription **302** paired with a corresponding non-synthetic speech representation of the corresponding transcribed speech utterance **304** in one or more training languages. Each of the one or more training languages is different from the target language. For example, the one or more training languages may include 52 languages with transcribed speech utterances and the target language may include another 50 languages (e.g., each different than the 52 training languages) with text-only training data. As will become apparent, the transcribed speech utterances **304** are used to train an alignment model **400** to generate alignment outputs **402** for the transcribed speech utterances **304** in the one or more training languages. Thereafter, the trained alignment model **400** is configured to receive, as input, the unspoken textual utterances **320** in the target language (e.g., different from each of the one or more training languages used to train the alignment model **400**) and generate, as output, alignment outputs **402** in the target language used to train the audio encoder **210**. Thus, training the alignment model **400** using the transcribed speech utterances **304** in the one or more training languages enables the alignment model **400** to generate alignment outputs **402** in the target language even though the alignment model **400** was never trained with any training data in the target language.

(22) For simplicity, the training process **300** includes a contrastive self-supervised loss part **300***a* (FIG. **3**A), a supervised loss part **300***b* (FIG. **3**B), and a consistency regularization part **300***c* (FIG.

3C). The training process **300** trains the audio encoder **210** on a total loss (L.sub.tts4pretrain2) based on: contrastive losses (L.sub.w2v) **316** derived using the contrastive self-supervised loss part **300***a* from the unspoken training text utterances (X.sub.text) **320** (e.g., in the target language), a corpus of transcribed non-synthetic speech utterances (X.sub.sup) **304** (e.g., in the one or more training languages), and un-transcribed non-synthetic speech utterances (X.sub.unsup) **306** (e.g., in the target language); supervised losses (L.sub.aux) **342**, **344** derived using the supervised loss part **300***b* from the unspoken training text utterances (X.sub.text) **320** and the transcribed non-synthetic speech utterances (X.sub.sup) **304**; and consistency losses (custom character.sub.cons (θ)) **352** derived using the consistency regularization part **300***c*.
(23) Referring to FIG. **3**A, the contrastive self-supervised loss part **300***a* of the training process **300** may employ an alignment model **400** that is configured to generate, at each of a plurality of output steps, alignment outputs (i.e., textual representations) **402** for each of a plurality of unspoken training text utterances **320**. The unspoken textual utterances **320** includes unspoken text that is text-only data in the target language, i.e., unpaired data, such that each unspoken textual utterance (X.sub.text) **320** is not paired with any synthesized or non-synthesized speech. Accordingly, the alignment model **400** generates a corresponding alignment output **402** for each of the unspoken textual utterances **320**.
(24) Referring now to FIG. **4**, in some examples, the alignment model **400** includes an embedding extractor **410**, duration predictor **420**, and an upsampler **430**. The embedding extractor **410** receives the unspoken textual utterance **320** that includes a sequence of text chunks including words, word-pieces, phonemes, and/or graphemes and extracts a corresponding initial textual representation (e.sub.t) **412**. The initial textual representation **412** includes embedding lexical information from the unspoken textual utterance **320**. Additionally or alternatively, the embedding extractor **410** may receive a transcription **302** corresponding to a transcribed non-synthetic speech utterance **304** (FIG. 3C). The duration predictor **420** receives the initial textual representation **412** from the embedding extractor **410** and predicts a corresponding text chunk duration (i.e., word, word-piece, phoneme, and/or grapheme duration) **422**. The text chunk duration **422** indicates a duration the corresponding text chunk would be spoken if a human (or text-to-speech system) spoke the unspoken textual utterance **320**. For example, the unspoken textual utterance **320** may include a sequence of phonemes and the duration predictor **420** predicts a phoneme duration **422** for each phoneme in the sequence of phonemes. In this example, the duration predictor **420** predicts the phoneme duration **422** by predicting a probability of non-zero duration for each phoneme and predicting a probability of continuous phoneme duration for each phoneme. As the sequence of phonemes includes regular phonemes, silences between word boundaries, and punctuation marks, only the regular phonemes are associated with non-zero duration while the silences and punctuation marks are generally associated with the continuous phoneme duration. Accordingly, the duration predictor **420** may use a sigmoid activation following a first one of two independent activations to predict the probability of non-zero duration and use a soft plus activation following a second one of the two independent projections to predict the continuous text chunk duration **422** for each text chunk. The duration predictor **420** determines, for each text chunk, whether the probability of non-zero duration is less than a threshold value, and when the probability of non-zero duration is less than the threshold value, a multiplier may zero-out the continuous text chunk duration **422** predicted by the softplus activation for the corresponding text chunk. Otherwise, when the probability of non-zero duration is not less than the threshold value, the predicted text chunk duration **422** may be set equal to the continuous phoneme duration predicted by the softplus activation.
(25) The upsampler **430** receives, for each unspoken textual utterance **320**, the corresponding initial textual representation **412** and the predicted text chunk duration **422**, and generates an alignment output (ê.sub.t) **402** having a number of frames by upsampling the initial textual representation **412** using the corresponding predicted text chunk duration **422**. Here, the alignment output **402** represents an aligned speech-text representation. In some examples, the alignment model **400** sends

the alignment output **402** to a text encoder **202** of the audio encoder **210** (FIGS. **3**B and **3**C). In other examples (not shown), the alignment model **400** sends the alignment output **402** to a shared encoder **250** (e.g., bypassing the text encoder **202**) of the audio encoder **210** (FIGS. **3**B and **3**C). In these other examples, the alignment output **402** serves as the encoded textual representation **312** such that the shared encoder **250** may receive the alignment output **402** directly from the alignment model **400**. In yet other examples, paired training data is available and the upsampler **430** generates the alignment output **402** as follows:

$$\hat{e}_t=\theta_{Refiner}(\text{Resample}(e_t,\text{Align}_{RNN\text{-}T}(e_s,t))) \qquad (1)$$

(26) Here, the upsampler **430** includes resampler and refiner layers that align the initial textual embedding **412** to align with a corresponding encoded audio representation **314** (FIGS. **3**B and **3**C) directly. However, when paired training data is not available, the upsampler **430** generates the alignment output **402** as follows:

$$\hat{e}_t=\theta_{Refiner}(\text{Resample}(e_t,\theta_{duration}(e_t))) \qquad (2)$$

(27) In particular, the number of frames of the alignment output **402** indicates a predicted speech duration of the unspoken textual utterance **320**. Stated differently, the number of frames of the alignment output **402** maps (i.e., aligns) the sequence of text chunks of the unspoken textual utterance **320** to speech frames. Here, the upsampler **430** includes resampler and refiner layers that replicate the initial textual embedding **412** to match the predicted text chunk duration **422** (i.e., speech duration). As such, the alignment output **402** includes a textual representation of the unspoken textual utterance **320** having a timing component that aligns with how a human would speak the unspoken textual utterance **320**. In some examples, the embedding extractor **410** receives the language identifier **321** that uniquely identifies the language of the one or more training languages and/or the target language to condition the alignment model **400**.

(28) In some implementations, training the alignment model **400** using training data in the one or more training languages, and then, generating alignment outputs **402** in the target language (e.g., different than each of the training languages) leads to low quality alignment outputs **402** because a script (e.g., Brahimic) of the target language has no overlap with scripts of the one or more training languages. To that end, in some examples, the alignments model **400** includes a pronunciation model that converts a script of the unspoken textual utterances **320** in the target language into a representation (e.g., phonetic representation) shared across multiple languages. In other examples, the alignment model **400** may transliterate the script of the unspoken textual utterances **320** into a different script. That is, the different script may align with the scripts of the transcribed speech utterances **304**.

(29) Notably, in most instances, a text-to-speech (TTS) system generates an audible output to give the unspoken textual utterance **320** the timing component of human speech such that a training process may use the audible output from the TTS system (i.e., synthetic speech) to train the audio encoder **210**. However, the alignment model **400** advantageously generates the alignment output **402** thereby mapping the sequence of text chunks to speech frames directly. As such, the training process **300** does not require any TTS system to generate synthetic speech from the unspoken textual utterances **320** to train the audio encoder **210**. That is, neither the training process **300** nor the alignment model **400** converts the unspoken textual utterance **320** into synthetic speech, but rather generates alignment outputs **402** (i.e., text alignments).

(30) FIG. **5** illustrates an example training process **500** for training the alignment model **400** using transcribed non-synthetic speech utterances **304** in the one or more training languages. That is, each of the one or more training languages is a different language than the target language audio encoder **210** (FIG. **3**) is trained to recognize. Moreover, each transcribed non-synthetic speech utterance **304** has a corresponding transcription **302**. In the example shown, the speech encoder **204** receives, as input, each transcribed non-synthetic speech utterance **304** as a sequence of features/vectors (e.g., mel-frequency spectrograms such as the acoustic frames **110** of FIG. **1**) and generates, as output, for each of a plurality of output steps, an encoded audio representation (e_s) **314** that

corresponds to the transcribed non-synthetic speech utterance **304** at the corresponding output step. In parallel, the alignment model **400** receives the transcription **302** corresponding to the same non-synthetic speech utterance **304** and generates an alignment output **402** according to Equation 1. The text encoder **202** receives, as input, the alignment output **402** and generates, as output, for each of the plurality of output steps, an encoded textual representation (ê.sub.t) **312** that corresponds to the transcription **302** at the corresponding output step.

(31) A modality loss module **550** receives the encoded textual representation **312** and the encoded audio representation **314** and generates a modality loss **552** based on comparing the encoded textual representation **3412** and the encoded audio representation as follows:

custom character.sub.MM=*MSE*(*e*.sub.s,*ê*.sub.t)+custom character.sub.RNNT(*t*|*e*.sub.s)
(3)

(32) Equation 3 adds the mean squared error (MSE) of the encoded textual representation (ê.sub.t) **312** and the encoded audio representation (e.sub.s) **314** to RNN-T model alignments between predicted text targets and the encoded audio representations (e.sub.s) **314** to determine the modality loss (custom character.sub.MM) **552**. Here, the encoded audio representations **314** serve as a ground-truth label to train the alignment model **400** to generate alignment outputs **402** that align to the corresponding non-synthetic speech utterances **304**. The training process **700** may use the modality loss **552** to update parameters of the alignment model **400**. For example, the training process **700** may update parameters of the duration predictor **420** and/or the upsampler **430** (FIG. **4**).

(33) FIG. **6** illustrates an example training process **600** for training the alignment model **400** using paired training data (i.e., transcribed speech utterances **304** in the one or more training languages) and unpaired training data (i.e., unspoken textual utterances **320** in the target language). That is, the training process **600** uses transcribed non-synthetic speech utterances **304** that have corresponding transcriptions **302** (i.e., paired training data) and unspoken textual utterances **320** (i.e., unpaired training data) to learn speech-aligned alignment outputs **402**. In the example shown, the speech encoder **204** receives, as input, each transcribed non-synthetic speech utterance **304** as a sequence of features/vectors (e.g., mel-frequency spectrograms such as the acoustic frames **110** of FIG. **1**) and generates, as output, for each of the plurality of output steps, an encoded audio representation (e.sub.s) **314** that corresponds to the transcribed non-synthetic speech utterance **304** at the corresponding output step. In parallel, the alignment model **400** receives the transcription **302** corresponding to the same non-synthetic speech utterance **304** and generates an alignment output according to Equation 1. Additionally or alternatively, the alignment model **400** may receive the unspoken textual utterance **320** and generate an alignment output **402** according to Equation 2. The text encoder **202** receives, as input, the alignment output **402** and generates, as output, for each of the plurality of output steps, an encoded textual representation (e.sub.t) **314**.

(34) The audio encoder **210** may include a shared encoder **250** that receives, as input, the encoded textual representations **312**, and generates, as output, a first encoded shared representation **322**. The shared encoder **250** may also receive, as input, the encoded audio representations **314** and generate, as output, a second encoded shared representation **324**. An auxiliary decoder **390** receives, as input, the first and second encoded shared representations **322**, **324** and generates, as output, corresponding first and second probability distributions **392**, **294** over possible speech recognition hypotheses.

(35) An alignment masked loss module **650** receives the first probability distribution **392** corresponding to the encoded textual representation **312** and the second probability distribution **394** corresponding to the encoded audio representation **314** and generates an alignment loss **652** as follows:

custom character.sub.A-MLM=custom character.sub.RNNT(*t*|Mask(*ê*.sub.t)) (4)

(36) The alignment loss **652** from Equation 4 may be applied over the masked, sampled encoded textual representations **312** in a frequency and time domain. Notably, the alignment loss **652** may

be used as a training objective for both paired training data and unpaired training data. The training process **600** may use the alignment loss **652** to update parameters of the alignment model **400**. For example, the training process **800** may update parameters of the duration predictor **420** and/or the upsampler **430** (FIG. **4**).

(37) Referring back to FIG. **3**A, in some implementations, the audio encoder **210** includes a speech encoder **204** and a text encoder **202**, described in more detail with reference to FIGS. **3**B and **3**C. In the example shown, the audio encoder **210** (alternatively the speech encoder **204** or the text encoder **202** (FIGS. **3**B and **3**C)) includes a Conformer encoder including a stack of Conformer blocks each of which includes a stack of multi-headed self-attention, depth wise convolution, and feed-forward layers. Alternatively, the audio encoder **210** may include another type of encoder having a stack of multi-head self-attention layers/blocks, such as a transformer or performer encoder. The Conformer encoder **210** can naturally be split into a feature encoder, including a convolution subsampling block **212**, and a context network, including a linear layer **214** and a stack of Conformer blocks **216**. In some implementations, the convolution subsampling block **212** has two two-dimensional-convolution layers, both with strides (2, 2), resulting in a 4× reduction in the feature sequence length. The convolution subsampling block **212** receives, as input, a sequence of input features/vectors (e.g., mel-frequency spectrograms such as the acoustic frames **110** of FIG. **1**) associated with each transcribed non-synthetic speech utterance **304** and each un-transcribed non-synthetic speech utterance **306**, and generates, as output, for each of a plurality of output steps, an encoded audio feature **211** that corresponds to a respective one of the transcribed non-synthetic speech utterances **304** or a respective one of the un-transcribed non-synthetic speech utterances **306**. The convolution subsampling block **212** may receive, as input, each alignment output **402** generated by the alignment model **400** from the unspoken textual utterances **320** and generate, as output, for each of the plurality of output steps, an encoded textual feature **213** that corresponds to a respective one of the alignment outputs **402**.

(38) The encoded audio and textual features **211**, **213** (i.e., interchangeably referred to as “encoded features **211**, **213**”) output from the convolution subsampling block **212** may be fed to a masking module **218** where some of the encoded features **211**, **213** are randomly chosen and replaced with a trained feature vector shared between all masked time steps to provide corresponding masked encoded audio features **211**, **211***m* and masked encoded textual features **213**, **213***m*. In some examples, the masking module **218** masks the randomly chosen encoded features **211**, **213** for masking by randomly sampling without replacement a certain proportion p of all time steps to be start indices and then masks the subsequent M consecutive time steps from every sample index, whereby some spans may overlap. After masking is applied, the linear layer **214** and the Conformer blocks **216** of the context network receive the masked encoded features **211***m*, **213***m* (or encoded features **211**, **213** not chosen by the masking module **218**) and outputs corresponding contrastive context vectors (i.e., encoded representation) **215** from masked encoded features **211***m*, **213***m*. Moreover, a quantizer **217** receives the encoded features **211**, **213** as input, and generates quantized vectors (i.e., target context vectors) **219** as output. Thereafter, a contrastive loss module **315** derives a contrastive loss (L.sub.w2v) **316** between the contrastive context vectors **215** at the masked positions and the target context vectors **219** as follows.

(39) $$\mathcal{L}_{w2v} = -\log \frac{\exp(\mathrm{sim}(c_t, q_t)/k)}{{}_{\tilde{q}\sim Q_t} \exp(\mathrm{sim}(c_t, \tilde{q})/k)} \quad (5)$$

(40) where c.sub.t is contrastive context vector **215** centered over a masked output step (i.e., time step) t and q.sub.t represents a target context vector **219** at the output step tin a set of K+1 candidate target context vectors **219** which includes q.sub.t and K distractors. Distractors may be uniformly sampled from other masked output steps of the same utterance.

(41) The contrastive loss **316** is optimized between the contrastive context vectors **215** at the masked positions and the target context vectors **219**. After the audio encoder **210** converges on the un-transcribed non-synthetic speech utterances **306**, the training procedure is repeated on both the

alignment outputs **402** corresponding to the unspoken textual utterance **320** and the transcribed non-synthetic speech utterances **304**. Thus, the contrastive loss **316** (L.sub.w2v) is optimized for both real/human (non-synthetic) and unspoken textual utterances **320** represented by alignment outputs **402**, with additional auxiliary losses derived from the transcribed non-synthetic speech utterances **304** and the alignment outputs **402** as described in greater detail below with reference to FIG. **3**B. Accordingly, the contrastive self-supervised loss part **300***a* of the training process **300** trains the audio encoder **210** using the contrastive loss **316** derived from the corresponding encoded features **211**, **213** associated with each alignment output **402**, each transcribed non-synthetic speech utterance **304**, and each un-transcribed non-synthetic speech utterance **306** provided as input to the audio encoder **210**. Training the audio encoder **210** may include updating parameters of the audio encoder **210** based on the contrastive losses **316**.
(42) Referring to FIG. **3**B, the supervised loss part **300***b* of the training process **300** is configured to inject lexical information into the audio encoder **210** during training based on supervised loss terms **342**, **344** derived from the transcribed non-synthetic speech utterances and the alignment outputs **402** corresponding to unspoken textual utterances **320** output by the alignment model **400**. Notably, the supervised loss part **300***b* leverages one or more auxiliary decoders **390** for generating the supervised loss terms **342**, **344**. The auxiliary decoders **390** may include Connectionist Temporal Classification (CTC) decoders, Listen Attend Spell (LAS) decoders, or RNN-T decoders. These auxiliary decoders **390** may include at least one of a phoneme decoder configured to decode a sequence of phonemes or a wordpiece decoder configured to decode a sequence of word pieces. The auxiliary decoders **390** could also include a grapheme decoder configured to decode a sequence of graphemes.
(43) During the supervised loss part **300***b*, the text encoder **202** of the audio encoder is configured to receive alignment outputs **402** (i.e., text embeddings) from the alignment model and the speech encoder is configured to receive transcribed non-synthetic speech utterances **304**. That is, the text encoder **202** of the audio encoder generates encoded textual representations **3123** for alignment outputs **402** (e.g., corresponding to an unspoken textual utterance **320**) and the speech encoder **204** of the audio encoder **210** generates encoded audio representations **314** for speech inputs (i.e., transcribed non-synthetic speech utterances **304**). Here, the encoded textual representations **312** and the encoded audio representations **314** may not both be compatible with the auxiliary decoders **390**. Thus, the audio encoder **210** may also include a shared encoder **250** that receives the encoded textual representations **312** as input, and generates a first encoded shared representation **322** (e.sub.text) as output. Moreover, the shared encoder **250** receives the encoded audio representations **314** as input, and generates a second encoded shared representation (e.sub.sup) **324** as output. Accordingly, the shared encoder **250** generates the first and second encoded shared representations **322**, **324** into a shared latent representation space compatible with the auxiliary decoder **390**.
(44) In particular, the shared encoder **250** receives, as input, each encoded textual representation **312** that corresponds to the alignment output **402** generated from the unspoken textual utterance **320** and generates, as output, for each of the plurality of output steps, the first encoded shared representation (e.sub.text) **322** that corresponds to the alignment output **402** at the corresponding output step. The auxiliary decoder **390** including the phoneme decoder, wordpiece decoder, or the byte decoder receives, as input, each first encoded shared representation **322** output from the shared encoder **250** and generates, as output, a first probability distribution **392** over possible speech recognition hypotheses for the corresponding alignment output **402** at the corresponding time step. In some examples, the first probability distribution **392** over possible speech recognition hypotheses includes one of possible phoneme labels, possible word piece labels, or possible grapheme labels. Thereafter, a supervised loss module **340** may determine an alignment output loss term **342** based on the first probability distribution **392** over possible speech recognition hypotheses for the alignment output **402** corresponding to the unspoken textual utterance **320**. Here, the corresponding unspoken textual utterance **320** in which the alignment output **402** is

generated from, also serves as a ground-truth transcription. The supervised loss part **300***b* may train the audio encoder **210** on the alignment output loss term **342** by updating parameters of the audio encoder **210** based on the alignment output loss term **342**.

(45) Similarly, during the supervised loss part **300***b*, the shared encoder **250** receives, as input, each transcribed encoded audio representation **314** that corresponds to the non-synthetic speech utterance **304** and generates, as output, for each of the plurality of output steps, a second encoded shared representation (e.sub.sup) **324** that corresponds to the transcribed non-synthetic speech utterance **304** at the corresponding output step. The auxiliary decoder **390** including the phoneme decoder, the wordpiece decoder, or the byte decoder receives, as input, each second encoded shared representation **324** output from the shared encoder **250** and generates, as output, a second probability distribution **394** over possible non-synthetic speech recognition hypotheses for the corresponding transcribed non-synthetic speech utterance **304** at the corresponding output step. In some examples, the second probability distribution **394** over possible non-synthetic speech recognition hypotheses includes the one of possible phoneme labels, the possible word piece labels, or the possible grapheme labels. Thereafter, the supervised loss module **340** may determine a non-synthetic speech loss term **344** based on the second probability distribution **394** over possible non-synthetic speech recognition hypotheses and the corresponding transcription **302** paired with the transcribed non-synthetic speech utterance **304**. Here, the corresponding transcription **302** serves as a ground-truth transcription and may include a sequence of target phonemes, target word pieces, and/or target graphemes. The supervised loss part **300***b* may train the audio encoder **210** on the non-synthetic speech loss term **344** by updating parameters of the audio encoder **210** based on the non-synthetic speech loss term **344**.

(46) In some implementations, the supervised loss part **300***b* of the training process **300** uses another auxiliary decoder **390** to generate a third probability distribution **393** over possible speech recognition hypotheses based on the first encoded shared representation (e.sub.text) **322** for the alignment output **402** at the corresponding output step, whereby the supervised loss module **340** may determine another alignment output loss term **342** based on the third probability distribution **393** and the unspoken textual utterance **320** corresponding to the alignment output **402**. Here, the other auxiliary decoder **390** includes the other one of the phoneme decoder, word piece decoder, or the grapheme decoder and the third probability distribution **393** over possible speech recognition hypotheses includes the other one of the possible phoneme labels, the possible word piece labels, or the possible grapheme labels. In these implementations, the other auxiliary decoder **390** also generates a fourth probability distribution **395** over possible non-synthetic speech recognition hypotheses for the corresponding second encoded shared representation **324** at the corresponding output step, whereby the supervised loss module **340** may determine another non-synthetic speech loss term **344** based on the fourth probability distribution **395** and the corresponding transcription **302** that is paired with the transcribed non-synthetic speech representation **304**. Here, the fourth probability distribution **395** over possible non-synthetic speech recognition hypotheses includes the other one of the possible phoneme labels, the possible word piece labels, or the possible grapheme labels. The supervised loss part **300***b* of the training process **300** may similarly the audio encoder **210** on the other alignment output loss term **342** and the other non-synthetic speech loss term **344**.

(47) The un-transcribed non-synthetic speech utterances **306** and the unspoken textual utterances **320** each correspond to "unpaired" training data whereby the contrastive loss (L.sub.w2v) **316** derived from the unspoken textual utterances (X.sub.text) **320** may be combined with the supervised loss custom character.sub.aux associated with the alignment output loss term **342** to obtain an unspoken textual loss function, custom character.sub.text, as follows.

custom character.sub.text=custom character.sub.w2v($x|\theta$.sub.e)+custom character .sub.aux($y|x,\theta$.sub.e,θ.sub.d) (6)

(48) Likewise, the contrastive loss (L.sub.w2v) **316** derived from the un-transcribed non-synthetic speech utterances (X.sub.unsuo) **306** may be used to express an unsupervised speech loss function,

custom character.sub.unsup_speech, as follows.

custom character.sub.unsup_speech=custom character.sub.w2v($x^*|\theta$.sub.e)   (7)

(49) During training of the audio encoder **210**, the alignment outputs **402** and the un-transcribed non-synthetic utterances **306** may be separated or mixed within each batch. In order to force the audio encoder **210** to learn representations that are effective for both alignment outputs **402** corresponding to unspoken textual utterances **320** and non-synthetic (human/real) speech, the loss mask σ is applied when combining the loss functions custom character.sub.text and of Equations 5 and 6 to obtain an unpaired data loss function, custom character.sub.unpaired, as follows:

custom character.sub.unpaired=σcustom character.sub.text+(1−σ)custom character .sub.speech   (8)

(50) The transcribed non-synthetic speech utterances **304** corresponds to "paired" and "supervised" training data whereby the derived contrastive loss L.sub.w2v and the derived supervised loss custom character.sub.aux associated with the non-synthetic speech loss term **344** may be combined to obtain a paired data loss function, custom character.sub.paired, as follows:

custom character.sub.paired=custom character.sub.w2v($x|\theta$.sub.e)+custom character .sub.aux($y|x,\theta$.sub.e,θ.sub.d)   (9)

(51) In some scenarios, after training the audio encoder **210**, the ASR model recognizes audio from the target language during inference with graphemes from the training languages. As such, in some implementations, the supervised part **300***b* of the training process **300** employs residual adaptor layers **330** that condition at least one of the audio encoder **210** or the decoder (e.g., prediction network **220** and joint network **230** (FIG. **2**)) on a language identifier **321** that uniquely identifies the target language. Each residual adaptor layer **330** includes a small feed forward network including self-attention layers (e.g., 2 self-attention layers) with a bottleneck dimension. Thus, the residual identifier output **332** is fed to the shared encoder **250** such that the audio encoder **210** is conditioned on the language identifier **321** and does not recognize audio from the target language during inference with graphemes from the training languages.

(52) Referring to FIG. **3**C, the consistency regularization part (i.e., modality matching part) **300***c* of the training process **300** is configured to promote the audio encoder **210** to learn consistent predictions between non-synthetic speech (e.g., real/human speech) and alignment outputs **402** corresponding to unspoken textual utterances **320** by generating a consistent loss term (custom character.sub.cons (θ)) **352** between training utterance pairs **301** that each include a corresponding one of the transcribed non-synthetic speech utterances (X.sub.sup) **304** and a paired alignment output **404** of the same utterance as the corresponding transcribed non-synthetic speech utterance **304**. As such, the non-synthetic speech utterance **304** and the paired alignment output **404** of each training utterance pair **301** is associated with a same ground-truth transcription. In short, the consistent loss term **352** between the transcribed non-synthetic speech utterance **304** and paired alignment output **404** of the same training utterance provides an unsupervised training aspect by encouraging the audio encoder **210** to behave consistently regardless of whether the training utterance belongs to non-synthetic speech (i.e., speech training data) or the alignment output (i.e., text training data) and independent of supervised loss terms between the ground-truth transcription **302** and each of: non-synthetic speech recognition hypotheses output by the auxiliary decoder **390**; and speech recognition hypotheses output by the auxiliary decoder **390**.

(53) Similar to the alignment outputs **402** generated from the unspoken textual utterances **320** in FIG. **3**B, the alignment model **400** may generate each paired alignment output **404** using the corresponding transcription **302** that is paired with transcribed non-synthetic speech utterance **304**. Here, the non-synthetic speech representation **304** is associated with paired alignment output **404** generated by the alignment model **400** mapping the unspoken textual utterance **320** into speech frames.

(54) During the consistency regularization part **300***c*, the text encoder **202** receives, as input, each paired alignment output **404** and generates, as output, for each of the plurality of output steps, an

encoded textual representation **313** that corresponds to the paired alignment output **404** at the corresponding output step. The shared encoder **250** receives, as input, the encoded textual representation **313** and generates, as output, a first encoded shared representation (e*.sub.sup) **323**. The auxiliary decoder **390** including the phoneme decoder or the wordpiece decoder receives, as input, each first encoded shared representation **323** output from the shared encoder **250** and generates, as output, a first probability distribution **311** over possible speech recognition hypotheses for the corresponding paired alignment output **404** at the corresponding output step. In some examples, the first probability distribution **311** over possible speech recognition hypotheses includes one of possible phoneme labels or possible word piece labels.

(55) Similarly, the speech encoder **204** receives, as input, each transcribed non-synthetic speech utterance **304** as a sequence of features/vectors (e.g., mel-frequency spectrograms such as the acoustic frames **110** of FIG. **1**) and generates, as output, for each of a plurality of output steps, a encoded audio representation **314** that corresponds to the transcribed non-synthetic speech utterance **304** at the corresponding output step. The shared encoder **250** receives, as input, the encoded audio representation **314** and generates, as output, a second encoded shared representation (e.sub.sup) **324**. The auxiliary decoder **390** including the phoneme decoder or the wordpiece decoder receives, as input, each second encoded shared representation **324** output from the shared encoder **250** and generates, as output, a second probability distribution **394** over possible non-synthetic speech recognition hypotheses for the corresponding transcribed non-synthetic speech utterance **304** at the corresponding time step. In some examples, the second probability distribution **394** over possible non-synthetic speech recognition hypotheses includes the one of the possible phoneme labels or the possible word piece labels.

(56) With continued reference to FIG. **3**C, the consistency regularization part **300***c* of the training process **300** further determines, at each of the plurality of time steps for each training utterance pair **301**, the consistent loss term (custom character.sub.cons (θ)) **352** for the corresponding training utterance pair **301** based on the first probability distribution **311** over possible speech recognition hypotheses and the second probability distribution **394** over possible non-synthetic speech recognition hypotheses. For instance, the training process **300** may employ a consistency loss term module **350** configured to receive, at each time step, the corresponding non-synthetic speech and speech recognition results **311**, **394** output by the auxiliary decoder **390**, and determine the consistency loss term **352** for the corresponding training utterance pair **301** at the time step.

(57) In some examples, the consistency regularization part **300***c* of the training process **300** determines the consistent loss term **352** based on a Kullback-Leibler divergence (D.sub.KL) between the first probability distribution **311** over possible speech recognition hypotheses and the second probability distribution **394** over possible non-synthetic speech recognition hypotheses. The consistent loss term **352** based on D.sub.KL may be expressed by the following equation:

custom character.sub.cons(θ)=custom character.sub.KL(*p*.sub.{tilde over (θ)}(*y*|*x*)‖*p*.sub.θ(*y*|*{circumflex over (x)}}*))   (10)

(58) Here, the consistent loss term **352** determined for the training utterance pair **301** at each time step provides an “unsupervised” loss term that is independent of the accuracy of the auxiliary decoder **390** (e.g., independent of the supervised loss terms **342**, **344** of FIG. **3**B), and thus, may be employed to update parameters of the audio encoder **210** for promoting consistency between non-synthetic speech representations and alignment outputs of the same utterances. In batch training, the consistent loss term **352** may correspond to an average loss term obtained for the batch. In other words, the consistent loss term **352** permits the audio encoder **210** to learn to behave the same, e.g., make consistent encoded representation predictions on both non-synthetic speech (e.g., real/human speech) and alignment outputs of a same training utterance, regardless of whether the training utterance belongs to non-synthetic speech or alignment outputs.

(59) Lastly, the training process **300** may combine the unpaired data loss function (custom character.sub.unpaired), the paired data loss function (custom character.sub.paired),

and the consistent loss term (custom character.sub.cons) to obtain an overall loss term, custom character.sub.tts4pretrain2, that may be expressed as follows.
custom character.sub.tts4pretrain2=custom character.sub.unpaired+λ.sub.1custom character paired+λ.sub.2custom character.sub.cons (11) where λ.sub.1 may be equal to 1.0 and λ.sub.2 is equal to 0.1. The training process **300** may pre-train the audio encoder **210** using the overall loss term, custom character.sub.tts4pretrain2, by updating parameters of the audio encoder **210** to effectively teach the audio encoder **210** to learn shared representations between speech and text in the target language even though no labeled training data in the target language is available. After training the audio encoder **210**, the training process **300** may fine-tune the pre-trained audio encoder on transcribed speech utterances that may include supervised training samples of both alignment outputs corresponding to unspoken textual utterance **320** and non-synthetic (e.g., human speech).

(60) In some implementations, the training process **300** for training the audio encoder **210** applies encoder consistency regularization. Unlike decoder consistency regularization applied to auxiliary decoder(s) during the consistency regularization part **300***c* that requires hypothesized labels (e.g., transcripts **302** and unspoken textual utterances **320**), encoder consistency regularization does not require hypothesized labels and therefore has the advantage being allowed to be applied to all the training data **304**, **306**, **320**. Encoder consistency regularization may be applied via Hierarchical Contrastive consistency Regularization (HCCR) techniques where encoder activations e, e* from original/non-augmented and augmented speech are projected through an auxiliary network to generate z and z*. Thereafter, positive and negative pairs are constructive and a contrastive loss l.sub.t,z,z* is calculated as follows:

(61) $$l_{t,z,z^*} = -\log \frac{\exp(\mathrm{sim}(z_t^*, z_t)/\quad)}{{}_{k=1}^{T}\exp(\mathrm{sim}(z_t^*, z_k)/\quad)} \qquad (12)$$

(62) Specific to HCCR, a Convolutional Neural Network (CNN) projection network may calculate projections over increasing length segments of encoder activations e (30, 50, 120 ms) to yield 3 views (V) and draw negative examples from the same utterance for short segments, and from other utterances in the batches with 120 ms segments. Accordingly, an HCCR loss may be calculated over the transcribed non-synthetic speech utterances **304** (paired speech), the un-transcribed non-synthetic speech utterances **306** (unpaired speech), and the alignment outputs **402** generated from the unspoken textual utterances **320** as follows:

(63) $$\mathcal{L}_{\mathrm{enc\_cons}} = \underset{v=1}{\overset{V}{.\mathrm{Math}.}} \ \underset{t=1}{\overset{T^{(v)}}{.\mathrm{Math}.}} \ l_{t,z^{*(v)},z^{(v)}} \qquad (13)$$

(64) The HCCR loss calculated by Equation 13 may be added to Equation 11 with a coefficient of 1 e−3 as part of the overall loss term, custom character.sub.tts4pretrain2, for use in pre-training the audio encoder **210**.

(65) Implementations described above describe the training process **300** training the training the audio encoder **210** for a target language, however, it is understood that the training process **300** may also be employed to train the audio encoder for multiple target languages each different from the one or more training languages. As such, the audio encoder **210** for a multilingual ASR model **200**. In some instances, the training process **300** may be employed to train end-to-end ASR models with decoder structures (i.e., non-pre-training) or fine-tune an ASR model to perform downstream tasks such as speech translation or natural language understanding. Moreover, implementations described above describe the training process using each part **300***a-c* of the training process **300**. Yet, it is understood any combination of the training parts **300***a-c* may be used to train the audio encoder **210** using any combination of unspoken textual utterances **320**, transcribed non-synthetic speech utterances **304**, and/or untranscribed non-synthetic speech utterances **306** independently.

(66) For instance, the transcribed non-synthetic speech utterances **304** in the one or more training languages may initially be used to train the alignment model **400** (FIGS. **5** and **6**). Thereafter, using the alignment model **400** trained using labeled training data in the one or more training languages,

the training process **300** may train the audio encoder using the alignment output loss term **342** derived from the unspoken textual utterances **320** in the target language(s) during the supervised loss part **300***b*. Advantageously, the training process **300** may leverage high-resource languages (e.g., where an abundance of labeled training data already exists) to initially train the alignment model **400** and use the trained alignment model **400** to train the audio encoder **210** on a low-resource (e.g., little or zero labeled training data exists) target language. Simply put, by training the alignment model using the large amount of labeled training data with one or more high-resource languages the alignment model **400** learns to generate alignment outputs in target languages even though the alignment model **400** was never trained on any data (or zero labeled training data) in the target language.

(67) In some scenarios, after training the audio encoder **210**, the ASR model recognizes audio from the target language during inference with graphemes from the training languages. As such, in some implementations, the consistency regularization part **300***c* of the training process **300** employs the residual adaptor layers **330** that condition at least one of the audio encoder **210** or the decoder (e.g., prediction network **220** and joint network **230** (FIG. **2**)) on the language identifier **321** that uniquely identifies the target language. Each residual adaptor layer **330** includes a small feed forward network including self-attention layers (e.g., 2 self-attention layers) with a bottleneck dimension. Thus, the residual identifier output **332** is fed to the shared encoder **250** such that the audio encoder **210** is conditioned on the language identifier **321** and does not recognize audio from the target language during inference with graphemes from the training languages.

(68) FIG. **7** is a flowchart of an example arrangement of operations for a computer-implemented method **700** of using aligned text and speech representations to train automatic speech recognition models without transcribed speech data. The method **700** may execute on data processing hardware **810** (FIG. **8**) using instructions stored on memory hardware **820** (FIG. **8**) may reside on the user device **102** and/or remote computer/server **201** of FIG. **1** corresponding to a computing device **800** (FIG. **8**).

(69) At operation **702**, the method **700** includes receiving training data including unspoken textual utterances **320** in a target language. Each unspoken textual utterance **320** not paired with any corresponding spoken utterance of non-synthetic speech (or synthetic speech). At operation **704**, the method **700** includes generating a corresponding alignment output **402** for each unspoken textual utterance **320** of the received training data using an alignment model **400**. The alignment model **400** is trained on transcribed speech utterances **304** in one or more training languages each different than the target language. That is, the alignment model **400** is trained on the training languages and generates the alignment outputs **402** for the unspoken textual utterances **320** in the target language that were unseen by the alignment model **400** during training. At operation **706**, the method **700** includes generating, using a text encoder **202**, a corresponding encoded textual representation **312** for each alignment output **402**. At operation **708**, the method **700** includes training a speech recognition model **200** on the encoded textual representation **312** generated for the alignment outputs **402** corresponding to the unspoken textual utterances **320** in the target language to teach the speech recognition model **200** to learn how to recognize speech in the target language. Notably, the only transcribed (i.e., paired) training data used to train the speech recognition model **200** to learn how to recognize speech in the target language is the transcribed speech utterances **304** in the one or more training languages, each of which is different than the target language, used to train the alignment model **400**.

(70) FIG. **8** is a schematic view of an example computing device **800** that may be used to implement the systems and methods described in this document. The computing device **800** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or

claimed in this document.

(71) The computing device **800** includes a processor **810**, memory **820**, a storage device **830**, a high-speed interface/controller **840** connecting to the memory **820** and high-speed expansion ports **850**, and a low speed interface/controller **860** connecting to a low speed bus **870** and a storage device **830**. Each of the components **810**, **820**, **830**, **840**, **850**, and **860**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **810** can process instructions for execution within the computing device **800**, including instructions stored in the memory **820** or on the storage device **830** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **880** coupled to high speed interface **840**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **800** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(72) The memory **820** stores information non-transitorily within the computing device **800**. The memory **820** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **820** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **800**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(73) The storage device **830** is capable of providing mass storage for the computing device **800**. In some implementations, the storage device **830** is a computer-readable medium. In various different implementations, the storage device **830** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **820**, the storage device **830**, or memory on processor **810**.

(74) The high speed controller **840** manages bandwidth-intensive operations for the computing device **800**, while the low speed controller **860** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **840** is coupled to the memory **820**, the display **880** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **850**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **860** is coupled to the storage device **830** and a low-speed expansion port **890**. The low-speed expansion port **890**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(75) The computing device **800** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **800***a* or multiple times in a group of such servers **800***a*, as a laptop computer **800***b*, or as part of a rack server system **800***c*.

(76) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof.

These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.
(77) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.
(78) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.
(79) To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.
(80) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method that when executed on data processing hardware causes the data processing hardware to perform operations comprising: receiving training data comprising unspoken textual utterances in a target language, each unspoken textual utterance not paired with any corresponding spoken utterance of non-synthetic speech; generating, using an alignment model, a corresponding alignment output for each unspoken textual utterance of the received training data, the alignment model trained on transcribed speech utterances in one or more training languages each different than the target language; generating, using a text encoder, a corresponding encoded textual representation for each alignment output; and training a speech recognition model on the encoded textual representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language to teach the speech recognition model to learn how to recognize speech in the target language.
2. The computer-implemented method of claim 1, wherein training the speech recognition model comprises training the speech recognition model without using any transcribed speech utterances in the target language for supervised learning.
3. The computer-implemented method of claim 1, wherein the speech recognition model comprises an audio encoder and a decoder.
4. The computer-implemented method of claim 3, wherein the decoder comprises a recurrent neural network transducer (RNN-T) architecture.
5. The computer-implemented method of claim 3, wherein the audio encoder comprises: the text encoder; a speech encoder; and a shared encoder.
6. The computer-implemented method of claim 3, wherein the audio encoder comprises a plurality of multi-headed self-attention layers.
7. The computer-implemented method of claim 3, wherein the audio encoder comprises a Conformer encoder.
8. The computer-implemented method of claim 3, wherein the operations further comprise conditioning at least one of the audio encoder or the decoder on a language identifier uniquely identifying the target language.
9. The computer-implemented method of claim 8, wherein conditioning the at least one of the audio encoder or the decoder comprises conditioning the at least one of the audio encoder or the decoder on the language identifier using residual adaptor layers.
10. The computer-implemented method of claim 1, wherein training the speech recognition model comprises: for each alignment output, generating, using a shared encoder, a first encoded shared representation of the alignment output in a shared latent representation space; for each transcribed speech utterance in the one or more training languages: determining, using a speech encoder, an encoded audio representation of the transcribed speech utterance; and generating, using the shared encoder, a second encoded shared representation of the transcribed speech utterance in the shared latent representation space, wherein training the speech recognition model comprises training the speech recognition model on the first encoded shared representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language and the second encoded shared representation generated for the transcribed speech utterances in the one or more training languages.
11. The computer-implemented method of claim 1, wherein each unspoken textual utterance comprises a sequence of words, word-pieces, graphemes, and/or phonemes.
12. The computer-implemented method of claim 1, wherein the operations further comprise converting, using a pronunciation model, a script of the unspoken textual utterances in the target language into a phonetic representation shared across multiple languages.
13. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising:

receiving training data comprising unspoken textual utterances in a target language, each unspoken textual utterance not paired with any corresponding spoken utterance of non-synthetic speech; generating, using an alignment model, a corresponding alignment output for each unspoken textual utterance of the received training data, the alignment model trained on transcribed speech utterances in one or more training languages each different than the target language; generating, using a text encoder, a corresponding encoded textual representation for each alignment output; and training a speech recognition model on the encoded textual representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language to teach the speech recognition model to learn how to recognize speech in the target language.
14. The system of claim 13, wherein training the speech recognition model comprises training the speech recognition model without using any transcribed speech utterances in the target language for supervised learning.
15. The system of claim 13, wherein the speech recognition model comprises an audio encoder and a decoder.
16. The system of claim 15, wherein the decoder comprises a recurrent neural network transducer (RNN-T) architecture.
17. The system of claim 15, wherein the audio encoder comprises: the text encoder; a speech encoder; and a shared encoder.
18. The system of claim 15, wherein the audio encoder comprises a plurality of multi-headed self-attention layers.
19. The system of claim 15, wherein the audio encoder comprises a Conformer encoder.
20. The system of claim 15, wherein the operations further comprise conditioning at least one of the audio encoder or the decoder on a language identifier uniquely identifying the target language.
21. The system of claim 20, wherein conditioning the at least one of the audio encoder or the decoder comprises conditioning the at least one of the audio encoder or the decoder on the language identifier using residual adaptor layers.
22. The system of claim 13, wherein training the speech recognition model comprises: for each alignment output, generating, using a shared encoder, a first encoded shared representation of the alignment output in a shared latent representation space; for each transcribed speech utterance in the one or more training languages: determining, using a speech encoder, an encoded audio representation of the transcribed speech utterance; and generating, using the shared encoder, a second encoded shared representation of the transcribed speech utterance in the shared latent representation space, wherein training the speech recognition model comprises training the speech recognition model on the first encoded shared representations generated for the alignment outputs corresponding to the unspoken textual utterances in the target language and the second encoded shared representation generated for the transcribed speech utterances in the one or more training languages.
23. The system of claim 13, wherein each unspoken textual utterance comprises a sequence of words, word-pieces, graphemes, and/or phonemes.
24. The system of claim 13 wherein the operations further comprise converting, using a pronunciation model, a script of the unspoken textual utterances in the target language into a phonetic representation shared across multiple languages.